UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

5090 BIOLOGY

5090/02 Paper 2 (Theory), maximum raw mark 80

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UNIVERSITY of CAMBRIDGE
International Examinations

1 **(a)** sepal(s) / calyx (R petal, A labels on diagram) ; [1]

(b) A / left hand flower + stamen(s) / anthers / pollen / androecium
(A negatives ref. female parts named) ; [1]

(c) **(i)** insect (no marks for **(ii)** if wrong, but mark **(ii)** if 'animal' or blank.)
(if they say self- or cross-pollination, A one mark max. for reasons which must refer to named sexual part) ;

mark the first, one per line – any **two** from:
filaments or stamens or anthers + short or protected AW,
stigma protected AW, large petals, (nectar) guides, stigma + lumpy / compact AW / not feathery AW, (landing) platform AW ;; [3]

(ii) mtf,opl, any two from : scent AW, colour / bright AW, nectar,
pollen grains rough or sticky ;; [2]

(d) cross-pollination or explanation ;

alleles / genes + (2) parents / plants ;

variation (in offspring) ;

colonisation of different habitats / wider distribution /
prevents overcrowding / less competition ;

adaptation / natural selection /evolution or described or named e.g.
(e.g. resistance) (R references to new species) ; [max 3]

2 **(a)** large surface area (to volume ratio) / long / thin ; [1]

(b) **D** villus/i (R microvillus)
(or fats for one mark) ; [1]

mtf,opl, any **two** from: amino acids, fatty acids, glycerol, glucose,
ions AW (or named e.g.), drugs/toxins, water, vitamins (or named e.g.), ;; [2]

E root hair (cell) ; [1]

(in either order) water ;
ions / salts / minerals / named ion / oxygen (R nitrogen) ; [2]

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(c) **(i)** feature drawn and labelled (A on extra diagram)
(R mitochondria / nucleus, but A reasons – see later) ; [1]

(ii) These must **both** be for the same feature for 2 marks, but if they do not match the label, then **0** for label.
(membrane) living ;
partially permeable AW ;
only selected molecules enter ;
allows osmosis or described (R diffusion) ;
active transport ; [max 2]

OR (ccw) permeable ;
tough / flexible ;
helps keep shape of cell ;
ref. part played in turgor AW ; [max 2]

OR (cell sap / vacuole / cytoplasm)
contains concentrated solution / ref. concentration gradient ;
for water entry ;
by osmosis / diffusion ; [max 2]

OR (for cytoplasm / mitochondria)
ref. respiration / energy release ;
for active transport ; [2]

OR (nucleus) – for **one** mark only
controls cell activites AW ; [1]
(R e.c.f. for villus)

3 **(a)** **(i)** more transpiration / water lost from lower surface or v.v. ; [1]

(ii) (A points anywhere)
waterproof AW ;
cuticle / (waxy) layer AW (must be ref. upper) (R skin) ;

correct ref. stomata / pores on upper or lower surface ;
release of <u>water</u> (vapour) (A only if stomata ref. has been made) ; [max 4]

(b) mtfopl, any **two** from:
paper / slide / leaf surface not properly dry, touched with
(wet) fingers, damaged vein / xylem in leaf, slow to set up expt.
after drying, any reasonable *qualified* student error ;; [2]

(c) run the experiment more times / repeat ;
take an average ;
perform expt. on another similar leaf (R leaf of different species) ; [max 1]

(d) Cobalt chloride paper on bench AW ;
timed (A takes less time to change) ; [2]

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4 **(a)** **(i)** carbohydrate / sugar / glucose / maltose / sucrose / starch ;

(ii) protein / amino acids (vitamins / ions etc.) ; [2]

(b) incubate / oven / suitable location ;
optimum / increased / suitable + temperature or stated (25 – 50 °C) ;
suitable pH ;
correct ref. enzymes ; [max 2]

(c) **(i)** H ; [1]

(ii) bacterium (or suitable, named e.g. *Streptococcus*) ;
antibiotics effective against bacteria AW ; [2]

(d) (A e.c.f. for virus)
bacteria or microorganisms mutate ;
become resistant to antibiotic (R immune) ;
antibiotic too readily prescribed / available ;
patients not completing course ; [max 3]

5 **(a)** **K** oesophagus / gullet ;
L duodenum / small intestine ; [2]

(b) starch – almost the same ;
protein – less ;
fibre – almost the same ; [3]
(A any symbols, but R more than one per line)

(c) **(i)** digested / changed / broken down / ref. enzyme action ;
to amino acids / (poly)peptides AW ;
absorption (available only with ref. amino acids) ; [max 2]

(ii) colon / rectum / large intestine ;
no suitable enzyme / not digested ;
ref. cellulose / lignin ;
therefore no absorption / egested as faeces / used in peristalsis
(A all other things, or named substances, are absorbed) ; [max 3]

[Maximum mark for Section A: 50]

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6 **(a)** **Q** (A so long as **Q or R** are mentioned – if no ref **Q** or **R**, max 2 for each)
eaten as <u>proteins</u> ;
ref. urine / faeces AW (I excretion refs. re. faeces) ;
urea ;
(ers)
decompos(ition) / decay / putrefaction or described ;
nitrification or described ;
correct ref. bacteria / fungi or named ;
<u>nitrates</u> + <u>proteins</u> + <u>plants</u> ;
(max 5 marks from above)

R carbon dioxide ;
from combustion AW ;
<u>respiration</u> (R if ref. only to oxygen) ;
photosynthesis + carbohydrates / named carbohydrate ;
(A on word equation)
[max 7]

(b) two from:
an oxide of nitrogen, SO_2, CO_2, CO, ;;

one from:
acid rain (for N_xO_y or SO_2 or CO_2) / global warming or greenhouse
effect AW (for CO_2 or CO) / CO + ref. carboxyhaemoglobin ; [3]

[Total: 10]

7 **(a)** (R any examples which do not come from the circulatory system, mark the first in a list.)

(i) blood cell / any correct named e.g. ;

(ii) muscle or correct named e.g. (A blood / capillary) ;

(iii) heart / blood vessel or named (R capillary) ; [3]

(b) <u>hepatic artery</u> (A name, disregard context) ;

carries more / most oxygen / oxygenated blood ;
more or ref. to being used or destroyed in the liver + insulin / glucagon /
adrenaline / hormones ;

<u>hepatic vein</u> (A name, disregard context) ;
carries carbon dioxide / deoxygenated (A with ref. HPV) ;
urea / ref. more amino acids in hepatic artery ;
any reasonable ref. to comparative numbers of RBCs + named blood vessel ;
(hepatic) <u>portal vein</u> / H<u>PV</u> (A name, disregard context) ;
#digested foods / any named ;
#ions / vitamins / water ;
any correct ref. to glucose levels in hepatic artery / vein ; [max 7]
(#both marks require a ref. somewhere to direction of blood flow)

[Total: 10]

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8E (a) externally administered substance AW / chemical ;
which modifies / changes / affects ;
metabolism / chemical reactions in the body AW ; [max 2]
(could be of bacteria)

(b) (alcohol) reduces self-control / removes inhibitions ;
<u>depressant</u> / ref. nervous system ;
slower reaction times or any named effect ;
liver damage / cirrhosis / other named condition ;
*expense ;
*social implications / crime ;
(heroin) <u>depressant</u> / ref. nervous system ;
*addiction / greater quantities for same effect / habituation / tolerance ;
withdrawal symptoms AW ;
needle-sharing ;
AIDS / hepatitis ; [max 8]
(Points marked * score *either* for alcohol *or* for heroin)

[Total: 10]

8O (a) <u>nicotine</u> ;
*craving / addiction / agglutination or clotting of RBCs / increased blood pressure / increased heart rate / vasoconstriction / increased blood fat / atheroma AW / arteriosclerosis / effect on fetus ;
(*All must be with ref. nicotine)

<u>tar</u> ;
decreased O_2 uptake (must be with ref. tar) ;
lung or any named correct form of cancer caused by tar ;

<u>CO</u> ;
decreased O_2 carriage (must be related to CO) ;

points not tied to any particular constituent:
emphysema / bronchitis / cough / destruction of cilia ;
heart disease / heart attack / stroke / atheroma (if not given above) ;
less O_2 to fetus ;
miscarriage / small / underweight babies (R underdeveloped) ; [max 8]

(b) smell / eye discomfort ;
ref. passive smoking or described ;
risk to health of non-smoker / ref. litter ; [max 2]

[Total: 10]

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